

Evaluate Expressions

Lesson 6-2

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.



x

stands for a number

In $3x + 5$, x can be any number — if $x = 2$, the expression equals 11; if $x = 10$, it equals 35

Variable

A letter that stands for an unknown number.



$3x + 5$

no equal sign

$2n + 7$ is an expression: the $2n$ means '2 times some number' and the $+ 7$ means 'then add 7'

Expression

A math phrase with numbers and letters, but no equal sign.



$3x + 5$

no equal sign

Evaluate $2n + 7$ when $n = 4$: replace n with $4 \rightarrow 2(4) + 7 = 8 + 7 = 15$

Evaluate

To find the value by putting numbers in for the letters.



$x \rightarrow 5$

put a number in for x

Substitute 3 for x in $5x$: replace x with 3 $\rightarrow 5(3) = 15$

Substitute

To put a number in place of a letter.

$$4x$$

coefficient = 4

*In $7m$, the coefficient 7 tells you to multiply m by 7
— if $m = 3$, then $7m = 21$*

Coefficient

The number in front of a letter, like the 3 in $3x$.

$$3x + 2x = 5x$$

same variable $\rightarrow$ combine

*$4x$ and $2x$ are like terms (both x^1); $4x$ and $4y$ are
NOT (different variables)*

Like terms

Terms with the same letter, like $2x$ and $5x$.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can evaluate algebraic expressions by substituting values for the variables.

1 Notice

You're a music producer setting up a concert.

2 Key idea

I can evaluate algebraic expressions by substituting values for the variables.

3 Words I'll use

Variable

Expression

Evaluate

Substitute

Coefficient

Like terms



Think About It

- What does the 150 represent in the expression?
- What does the 12s represent?
- What happens to the total cost as s gets bigger?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: Evaluate $3x + 7$ when $x = 4$.

- A. 19
- B. 34
- C. 12
- D. 14

Step 1

$$3(4) + 7 = 12 + 7 = 19.$$



Answer: A. 19

 We do — together

Solve this one with your class using the same steps.

Problem: Evaluate $20 - 2n$ when $n = 6$.

- A. 8
- B. 14
- C. 32
- D. 12

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: Which expression means '5 less than twice a number n '?

- A. $2n - 5$
- B. $5 - 2n$
- C. $2(n - 5)$
- D. $2 - 5n$

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The concert cost is $150 + 12s$, where s is the number of speakers. How do you find the cost for 5 speakers, and why does this one expression work for any number of speakers?

Level 1 support

Start here: Substitute the number of speakers for s , then evaluate.

 **Need a hint?**

Sentence starters (optional)

I substitute $s = 5$ into $150 + 12s$ to get ____.

Sustituyo $s = 5$ en $150 + 12s$ para obtener ____.

One expression works for any show because s ____.

Una expresión sirve para cualquier concierto porque s ____.

Word bank: Variable Expression Evaluate Substitute

Listen for: Students substitute $s = 5$ ($150 + 12 \times 5 = 210$), explain 150 is the fixed venue fee and 12 is the cost per speaker, and note s can be any number.

Level 2

Predict whether doubling the speakers from 5 to 10 doubles the total cost. Justify using the \$150 base fee.

- Doubling speakers does ____ double the cost because the 150 ____.
- For 10 speakers the cost is ____, which compared to 210 shows ____.

EXPLORE

When you evaluate an expression like $4x^2 + 3$, what order do you follow, and why must you square before you multiply?

Level 1 support

Start here: Follow order of operations: substitute, then exponents, then multiply, then add.

 **Need a hint?**

Sentence starters (optional)

First I substitute, then I do ____ because of the order of operations.

Primero sustituyo, luego hago ____ por el orden de las operaciones.

I must square before multiplying because ____.

Debo elevar al cuadrado antes de multiplicar porque ____.

Word bank: **Evaluate** **Substitute** **Expression** **Variable**

Listen for: A strong answer substitutes the value first, then applies the exponent before multiplying (e.g., $4 \times 2^2 = 4 \times 4 = 16$, not $(4 \times 2)^2$), citing order of operations.

Level 2

A classmate evaluates $4x^2 + 3$ at $x = 2$ as $(4 \times 2)^2 + 3 = 67$. Critique the error and give the correct value.

- The error is ____ because the exponent applies only to ____.
- The correct value is ____ because I do $2^2 =$ ____ first, then ____.

CONNECT

The rideshare fare is $2.50 + 1.75m$ for m miles. How much is an 8-mile ride, and what does each part of the expression represent?

Level 1 support

Start here: Substitute $m = 8$ and evaluate to get the total fare.

 **Need a hint?**

Sentence starters (optional)

I substitute $m = 8$: $2.50 + 1.75 \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$.

Sustituyo $m = 8$: $2.50 + 1.75 \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$.

The 2.50 is the $\underline{\hspace{1cm}}$ and the 1.75 is the $\underline{\hspace{1cm}}$.

El 2.50 es la $\underline{\hspace{1cm}}$ y el 1.75 es la $\underline{\hspace{1cm}}$.

Word bank: Substitute Variable Evaluate Expression

Listen for: Students evaluate $2.50 + 1.75 \times 8 = 16.50$ and identify 2.50 as the base fee and 1.75 as the per-mile rate.

Level 2

Two riders pay the same fare, but one traveled fewer miles. How is that possible with $2.50 + 1.75m$? Justify what would have to differ.

- That is possible only if $\underline{\hspace{1cm}}$, because the per-mile rate 1.75 $\underline{\hspace{1cm}}$.
- The base fee 2.50 is the same for both, so any difference comes from $\underline{\hspace{1cm}}$.

Write About the Math

The Writing Revolution

I can explain my steps using the words variable, expression, evaluate, and substitute.

1. Kernel Sentence subject + verb

Model: Expression is a math phrase with numbers and letters, but no equal sign.

Expresión es una frase matemática con números y letras, pero sin signo igual.

Write a kernel sentence about expression. Use a subject and a verb.

Escribe una oración base sobre expresión. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Expression matters in math

Expresión importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Expression matters in math because ____.

Expresión importa en matemáticas porque ____.

but
pero

Expression matters in math, but ____.

Expresión importa en matemáticas, pero ____.

so
entonces

Expression matters in math, so ____.

Expresión importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about expression.
Di un hecho verdadero sobre expression.

Expression ____.

Question
Pregunta

Ask a question about expression.
Haz una pregunta sobre expression.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about expression.
Muestra entusiasmo sobre expression.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with expression.
Dile a un compañero qué hacer con expression.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

I substitute $s = 5$ into $150 + 12s$ to get ____.

Sustituyo $s = 5$ en $150 + 12s$ para obtener ____.

One expression works for any show because s ____.

Una expresión sirve para cualquier concierto porque s ____.

First I substitute, then I do ____ **because of the order of operations.**

Primero sustituyo, luego hago ____ *por el orden de las operaciones.*

Try It

Solve on your own. Check the answer key when you are done.

1. If $a = 4$, what is $2a + 5$?

- A. 13
- B. 9
- C. 20
- D. 11

Show your work:

2. Evaluate $24 \div (2 + 4) \times 3$.

- A. 12
- B. 1.33
- C. 4
- D. 18

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A gym membership costs \$25 per month plus a \$40 signup fee. Write an expression using m for months, then evaluate it for 6 months and 12 months. Explain what each part of the expression represents.

Sentence starter: The expression is _____. For 6 months: _____ = _____. For 12 months: _____ = _____. The _____ represents the _____ and the _____ represents the _____.

Show your work:

Reflect — Exit Ticket

Evaluate $5(n - 3) + 2$ when $n = 7$.

- A. 22
- B. 32
- C. 17
- D. 36

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. $8 - 20 - 2(6) = 20 - 12 = 8$.
2. **You Try (notes):** A. $2n - 5$ — *Twice n is $2n$. Five less than that is $2n - 5$.*
3. **Try It 1:** A. $13 - 2 \times 4 + 5 = 8 + 5 = 13$.
4. **Try It 2:** A. $12 - 2 + 4 = 6$; $24 \div 6 = 4$; $4 \times 3 = 12$.
5. **Exit Ticket:** A. $22 - 5(7 - 3) + 2 = 5(4) + 2 = 20 + 2 = 22$.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Expression is a math phrase with numbers and letters, but no equal sign.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).